Lab 7: Trajectories and Moving Objects

Computer Graphics Lab, Wolfram Mathematica

In[1]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True]
SetOptions[{Plot, Plot3D, ContourPlot, ContourPlot3D, DensityPlot,
   ParametricPlot3D, ParametricPlot, ListPlot, ListLinePlot, ListPointPlot3D,
   Graphics, Graphics3D, PolarPlot}, ImageSize → Small, AxesLabel → {"x", "y", "z"}];
```

In[3]:=
```
(* Problem 1.1 *)
```

In[4]:=
```
ListN01 = {1, 2, -5, 8, 10, 22, -5, -9, 28, 5, 6, 12, 17}
```

Out[4]=
```
{1, 2, -5, 8, 10, 22, -5, -9, 28, 5, 6, 12, 17}
```

In[5]:=
```
ListN02 = {1, 1, 1, 1, 1, -2, -5, -9, -28, -5, -6, -12, 10}
```

Out[5]=
```
{1, 1, 1, 1, 1, -2, -5, -9, -28, -5, -6, -12, 10}
```

In[6]:=
```
PosN[L_] := Module[{S, i, T},
  T = Table[If[L[[i]] > 0, L[[i]], Nothing], {i, Length[L]}];
  S = Sum[T[[i]], {i, Length[T]}];
  S]
```

In[7]:=
```
PosN[ListN01]
```

Out[7]=
```
111
```

In[8]:=
```
PosN[ListN02]
```

Out[8]=
```
15
```

In[9]:=
```
(* Problem 1.2 *)
```

In[10]:=
```
ListN3 = {0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 2}
```

Out[10]=
```
{0, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 2}
```

In[11]:=
```
Aver1[L_] := Module[{S1, i, T},
  T = Table[If[L[[i]] ≠ 0, 1 / L[[i]], Nothing], {i, Length[L]}];
  S1 = Sum[T[[i]], {i, Length[T]}];
  {T, S1}]
```

In[12]:=
```
Aver1[ListN3]
```

Out[12]=

$$\left\{\left\{1, 1, 1, \frac{1}{2}\right\}, \frac{7}{2}\right\}$$

In[13]:=
```
(* Problem 2.1 *)
```

In[14]:=
```
PrN1N2[N1_, N2_] := Module[{T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, N1, N2, 1}];
  T]
```

In[15]:=
```
LP = PrN1N2[10, 50]
```

Out[15]=

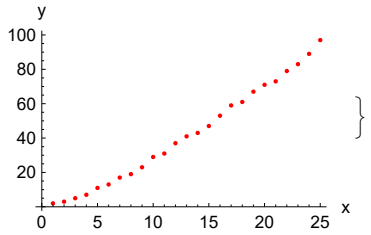
```
{11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47}
```

In[16]:=
```
PrN1N2Plot[N1_, N2_] := Module[{i, plot, T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, N1, N2, 1}];
  plot = ListPlot[T, PlotStyle → {Red, PointSize[0.015]}];
  {T, plot}]
```

In[17]:=
```
PrN1N2Plot[1, 100]
```

Out[17]=

```
{{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97},
```



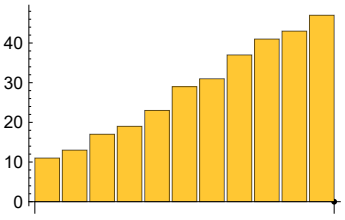

```
}
```

In[18]:=
```
PrN1N2Bar[N1_, N2_] := Module[{i, plot, T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, N1, N2, 1}];
  plot = BarChart[T];
  {T, plot}]
```

In[19]:=
```
PrN1N2Bar[10, 50]
```

Out[19]=

```
{{11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47},
```




```
}
```

In[20]:=
```
(* Plot 2.2 *)
```

In[21]:=
```
fN[k_] := Module[{i, T},
  T = Table[If[PrimeQ[i], i, Nothing], {i, 1, k, 1}];
  Length[T]]
```

In[22]:=
```
fN[50]
```

Out[22]=

15

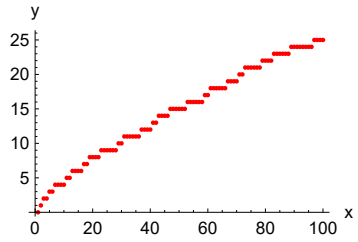
In[23]:=
```
LN = Table[{i, fN[i]}, {i, 1, 100}]
```

Out[23]=

```
{{1, 0}, {2, 1}, {3, 2}, {4, 2}, {5, 3}, {6, 3}, {7, 4}, {8, 4}, {9, 4}, {10, 4},
 {11, 5}, {12, 5}, {13, 6}, {14, 6}, {15, 6}, {16, 6}, {17, 7}, {18, 7}, {19, 8},
 {20, 8}, {21, 8}, {22, 8}, {23, 9}, {24, 9}, {25, 9}, {26, 9}, {27, 9}, {28, 9},
 {29, 10}, {30, 10}, {31, 11}, {32, 11}, {33, 11}, {34, 11}, {35, 11}, {36, 11},
 {37, 12}, {38, 12}, {39, 12}, {40, 12}, {41, 13}, {42, 13}, {43, 14}, {44, 14},
 {45, 14}, {46, 14}, {47, 15}, {48, 15}, {49, 15}, {50, 15}, {51, 15}, {52, 15},
 {53, 16}, {54, 16}, {55, 16}, {56, 16}, {57, 16}, {58, 16}, {59, 17}, {60, 17},
 {61, 18}, {62, 18}, {63, 18}, {64, 18}, {65, 18}, {66, 18}, {67, 19}, {68, 19},
 {69, 19}, {70, 19}, {71, 20}, {72, 20}, {73, 21}, {74, 21}, {75, 21}, {76, 21},
 {77, 21}, {78, 21}, {79, 22}, {80, 22}, {81, 22}, {82, 22}, {83, 23}, {84, 23},
 {85, 23}, {86, 23}, {87, 23}, {88, 23}, {89, 24}, {90, 24}, {91, 24}, {92, 24},
 {93, 24}, {94, 24}, {95, 24}, {96, 24}, {97, 25}, {98, 25}, {99, 25}, {100, 25}}
```

In[24]:=
```
GfN = ListPlot[LN, PlotStyle → {Red, PointSize[0.015]}]
```

Out[24]=



In[25]:=
```
InT = LN[[{1, 25, 50, 75, 100}]]
```

Out[25]=

```
{{1, 0}, {25, 9}, {50, 15}, {75, 21}, {100, 25}}
```

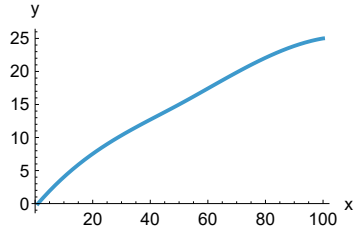
In[26]:=
```
fQint = Interpolation[InT, Method → "Spline"]
```

Out[26]=

InterpolatingFunction [Domain: {{1, 100}} Output: scalar]

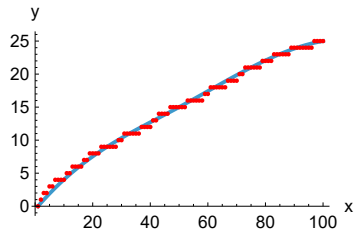
In[27]:=
```
GF = Plot[fQint[t], {t, 1, 100}]
```

Out[27]=



In[28]:=
```
Show[{GF, GfN}]
```

Out[28]=



In[29]:=
```
(* Problem 3 *)
```

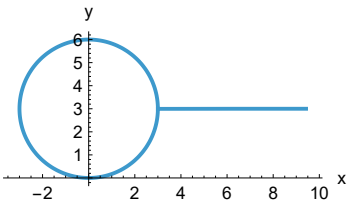
In[30]:=
```
R1 = 3
```

Out[30]=
```
3
```

In[31]:=
```
f1[t_] := {R1 * Cos[t], R1 * Sin[t] + R1}
f2[t_] := {t, R1}
Tr1[t_] := If[t ≥ 0 && t ≤ 2 * Pi, f1[t], f2[t]]
```

In[34]:=
```
ParametricPlot[Tr1[t], {t, 0, 3 Pi}, Exclusions → None]
```

Out[34]=



In[35]:=
```
F[x_] := Which[x > 0, 1, x < 0, 2, x == 0, 3]
```

In[36]:=
```
F[1]
```

Out[36]=
```
1
```

In[37]:=
```
F[-2]
```

Out[37]=
```
2
```

In[38]:=
```
F[0]
```

Out[38]=
```
3
```

In[39]:=
```
F[x_] := Which[x < 0, f1[x], x ≤ 5 && x ≥ 0, f1[x], x > 5, f1[x]]
```

In[40]:=
```
F[1]
```

Out[40]=
```
{3 Cos[1], 3 + 3 Sin[1]}
```

In[41]:=
```
(* Problem 4 *)
```

In[42]:=
```
Clear[t, f1, f2, f3, Tra5, Tri5]
```

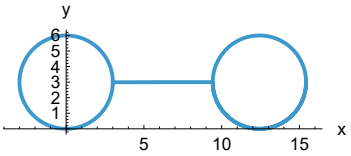
In[43]:=
```
R11 = 3;
```

In[44]:=
```
f1[t_] := {R11 * Cos[t], R11 * Sin[t] + R11}
f2[t_] := {t, R11}
f3[t_] := {R11 * Cos[t] + R11 + 3 * Pi, R11 * Sin[t] + R11}
```

In[47]:=
```
Tra5[t_] := Which[t ≥ 0 && t ≤ 2 * Pi, f1[t], t > 2 * Pi && t < 3 * Pi, f2[t], t ≥ 3 * Pi, f3[t]]
```

In[48]:=
```
ff = ParametricPlot[Tra5[t], {t, 0, 6 * Pi}, Exclusions → None]
```

Out[48]=



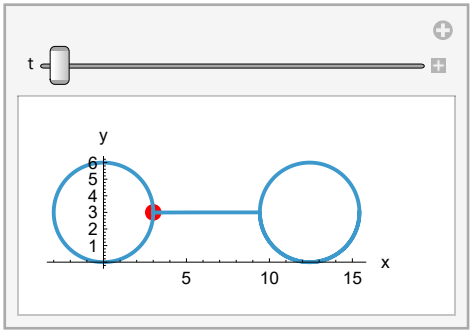
In[49]:=
```
(* Problem 5 *)
```

In[50]:=
```
Tri5[t_] := Graphics[{Red, Disk[Tra5[t], 1 / 2]}, Axes → True]
```

In[51]:=
```
ShowAll[t_] := Show[Tri5[t], ff]
```

In[52]:=
```
Manipulate[ShowAll[t], {t, 0, 6 * Pi}]
```

Out[52]=



In[53]:=
```
(* Problem 6 *)
```

In[54]:=
```
Ladder = {{1, 1}, {5, 5}, {9, 9}, {13, 13},
  {17, 17}, {21, 21}, {24, 21}, {28, 17}, {32, 13}, {36, 9}, {40, 5}}
```

Out[54]=
```
{{1, 1}, {5, 5}, {9, 9}, {13, 13}, {17, 17},
 {21, 21}, {24, 21}, {28, 17}, {32, 13}, {36, 9}, {40, 5}}
```

In[55]:=
```
tmax = Max[Transpose[Ladder][[1]]]
```

Out[55]=
```
40
```

In[56]:=
```
fLadder = Interpolation[Ladder, InterpolationOrder → 0]
```

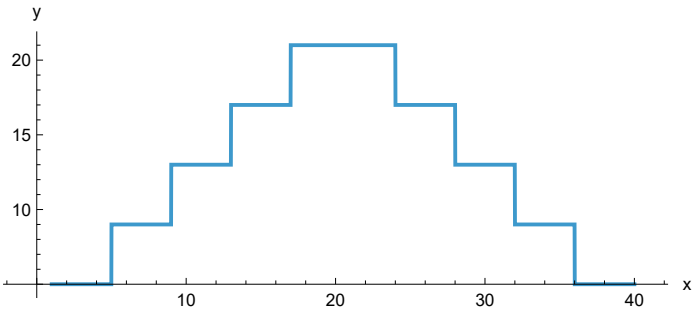
Out[56]=

InterpolatingFunction[Domain: {{1, 40}} Output: scalar]

In[57]:=
```
fL5[t_] := {t, fLadder[t]}
```

In[58]:=
```
Tra15 = ParametricPlot[fL5[t], {t, 1, tmax}, ImageSize → Medium]
```

Out[58]=



In[59]:=
```
nreg = 3; RR5 = 5;
```

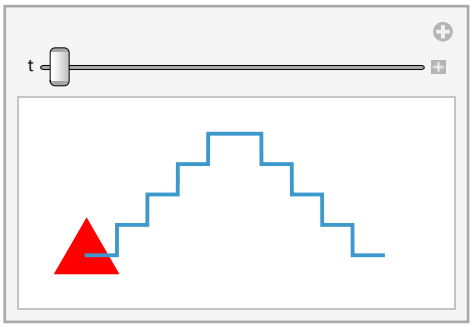
In[60]:=
```
ObjectPol[s_] := {Red, RegularPolygon[fL5[s], RR5, nreg]}
```

In[61]:=
```
Tri[s_] := Graphics[ObjectPol[s]]
```

In[62]:=
```
ShowAll2[s_] := Show[{Tri[s], Tra15}]
```

In[63]:=
```
Manipulate[ShowAll2[t], {t, 1, tmax}]
```

Out[63]=



In[64]:=
```
(* Problem 7 *)
```

In[65]:=
```
M7[nreg_, Rreg_] := Module[{ObjectPol, GP, ShowAll5, s},
  ObjectPol[s_] := {Red, RegularPolygon[fL5[s], RR5, nreg]};
  GP[s_] := Graphics[ObjectPol[s]];
  ShowAll5[t_] := Show[{Tra15, GP[t]}];
  ShowAll5]
```

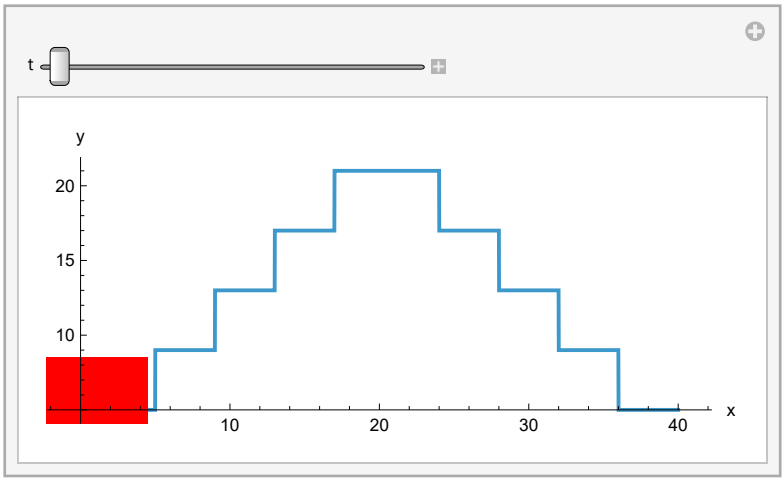
In[66]:=
```
f7 = M7[4, 2]
```

Out[66]=

```
ShowAll5$11194
```

In[67]:=
```
Manipulate[f7[t], {t, 1, tmax}]
```

Out[67]=



In[68]:=
```
(* Problem 8 *)
```

In[69]:=
```
Clear[GPnew, GGPnew]
```

In[70]:=
```
ObjectNew = {Red, RegularPolygon[{0, 0}, RR5, nreg]}
```

Out[70]=

```
{■, RegularPolygon[{0, 0}, 5, 3]}
```

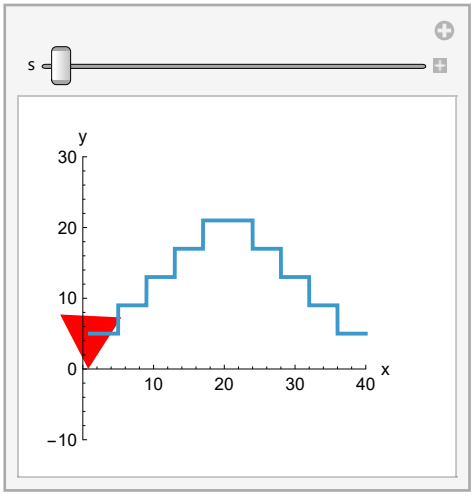
In[71]:=
```
ObjectRot[a_, s_] := GeometricTransformation[ObjectNew,
  Composition[TranslationTransform[fL5[s]], RotationTransform[a]]]
```

In[72]:=
```
GPnew[a_, s_] := Graphics[ObjectRot[a, s], PlotRange → {{0, 40}, {-10, 30}}, Axes → True]
```

In[73]:=
```
GGPnew[a_, s_] := Show[GPnew[a, s], Tra15]
```

In[74]:=
```
Manipulate[GGPnew[s, s], {s, 1, tmax}]
```

Out[74]=



In[75]:=
```
(* Problem 9 *)
```

In[76]:=
```
data9 = {{6.25, 0.0}, {6.001765667929582`, 0.4991670832341408`},
    {5.278004317051997`, 0.9933466539753061`}, {4.13971935999042`, 1.477601033306698`},
    {2.6826721590673235`, 1.9470917115432527`}, {1.0290638906736893`, 2.397127693021015`},
    {-0.6830632803695947`, 2.823212366975177`},
    {-2.3117715943739867`, 3.2210884361884555`},
    {-3.7234679805828397`, 3.586780454497614`}, {-4.804363328451665`, 3.916634548137417`},
    {-5.4701460286861545`, 4.207354924039483`}, {-5.6730252461135064`, 4.456036800307177`},
    {-5.4055342260972905`, 4.660195429836132`}, {-4.700772462711884`, 4.817790927085965`},
    {-3.62908203253932`, 4.927248649942301`}, {-2.2914696179044554`, 4.987474933020272`},
    {-0.810373552546205`, 4.997868015207525`}, {0.6813909728405793`, 4.958324052262343`},
    {2.05251635929549`, 4.869238154390976`}, {3.1848542843027783`, 4.731500438437072`},
    {3.983796907172315`, 4.546487134128409`}, {4.386467155241201`, 4.316046833244369`},
    {4.366996872319622`, 4.0424820190979505`}, {3.938434841957967`, 3.7285260608836`},
    {3.1511303019605785`, 3.377315902755753`}, {2.087754321004162`, 2.992360720519782`},
    {0.8554229419387158`, 2.577506859107321`}, {-0.42435467375067093`, 2.136899401169149`},
    {-1.6269859224456151`, 1.6749407507795233`},
    {-2.63633383181035`, 1.19624664606991`}, {-3.3554384511104276`, 0.7056000402993361`},
    {-3.715449005983995`, 0.20790331216645247`}, {-3.681958130522893`,
     -0.2918707171379004`}, {-3.258165023178643`, -0.7887284707162433`},
    {-2.4845813971987143`, -1.2777055101341583`}, {-1.435306822050832`,
     -1.7539161384480992`}, {-0.21121038911135578`, -2.2126022164742625`},
    {1.0693642937964811`, -2.649180704542467`}, {2.281529134967273`, -3.0592894547135963`},
    {3.3054259689726977`, -3.4388307959198707`},
    {4.037394751401694`, -3.7840124765396412`}, {4.399719513203253`, -4.091385555322054`},
    {4.348062109859167`, -4.357878862067941`}, {3.8759001316492907`, -4.580829683747274`},
    {3.0155535206251787`, -4.758010369447581`}, {1.8356904891936137`, -4.887650588325485`},
    {0.43551986299002976`, -4.968455018167322`},
    {-1.0638243785898849`, -4.999616287820504`}, {-2.529946600811446`,
     -4.9808230441792025`}, {-3.8309547730765376`, -4.912263063121662`},
    {-4.847278975639672`, -4.794621373315692`}, {-5.482485967105465`, -4.629073411638661`},
    {-5.672118447703113`, -4.417273278600765`}, {-5.389760456496705`, -4.161337211119504`},
    {-4.649774834880334`, -3.8638224377799357`}, {-3.5064531650285042`,
     -3.5277016278519597`}, {-2.0496368572833727`, -3.1563331893616042`},
    {-0.39718177268071464`, -2.753427712988188`}, {1.3150800368523572`,
     -2.323010897068783`}, {2.9450042957329634`, -1.8693833241511801`},
    {4.3564009896360165`, -1.3970774909946293`}, {5.4308248157476635`,
     -0.9108125213604751`}, {6.07785134254383`, -0.41544701408748197`}};
```

In[77]:=
```
data91 = Join[data9, {data9[[1]]}]
```

Out[77]=
```
{{6.25, 0.}, {6.00177, 0.499167}, {5.278, 0.993347}, {4.13972, 1.4776},
 {2.68267, 1.94709}, {1.02906, 2.39713}, {-0.683063, 2.82321}, {-2.31177, 3.22109},
 {-3.72347, 3.58678}, {-4.80436, 3.91663}, {-5.47015, 4.20735}, {-5.67303, 4.45604},
 {-5.40553, 4.6602}, {-4.70077, 4.81779}, {-3.62908, 4.92725}, {-2.29147, 4.98747},
 {-0.810374, 4.99787}, {0.681391, 4.95832}, {2.05252, 4.86924}, {3.18485, 4.7315},
 {3.9838, 4.54649}, {4.38647, 4.31605}, {4.367, 4.04248}, {3.93843, 3.72853},
 {3.15113, 3.37732}, {2.08775, 2.99236}, {0.855423, 2.57751}, {-0.424355, 2.1369},
 {-1.62699, 1.67494}, {-2.63633, 1.19625}, {-3.35544, 0.7056}, {-3.71545, 0.207903},
 {-3.68196, -0.291871}, {-3.25817, -0.788728}, {-2.48458, -1.27771}, {-1.43531, -1.75392},
 {-0.21121, -2.2126}, {1.06936, -2.64918}, {2.28153, -3.05929}, {3.30543, -3.43883},
 {4.03739, -3.78401}, {4.39972, -4.09139}, {4.34806, -4.35788}, {3.8759, -4.58083},
 {3.01555, -4.75801}, {1.83569, -4.88765}, {0.43552, -4.96846}, {-1.06382, -4.99962},
 {-2.52995, -4.98082}, {-3.83095, -4.91226}, {-4.84728, -4.79462}, {-5.48249, -4.62907},
 {-5.67212, -4.41727}, {-5.38976, -4.16134}, {-4.64977, -3.86382}, {-3.50645, -3.5277},
 {-2.04964, -3.15633}, {-0.397182, -2.75343}, {1.31508, -2.32301}, {2.945, -1.86938},
 {4.3564, -1.39708}, {5.43082, -0.910813}, {6.07785, -0.415447}, {6.25, 0.}}
```

In[78]:=
```
x9 = Transpose[data91][[1]]
```

Out[78]=
```
{6.25, 6.00177, 5.278, 4.13972, 2.68267, 1.02906, -0.683063, -2.31177, -3.72347, -4.80436,
 -5.47015, -5.67303, -5.40553, -4.70077, -3.62908, -2.29147, -0.810374, 0.681391,
 2.05252, 3.18485, 3.9838, 4.38647, 4.367, 3.93843, 3.15113, 2.08775, 0.855423, -0.424355,
 -1.62699, -2.63633, -3.35544, -3.71545, -3.68196, -3.25817, -2.48458, -1.43531,
 -0.21121, 1.06936, 2.28153, 3.30543, 4.03739, 4.39972, 4.34806, 3.8759, 3.01555,
 1.83569, 0.43552, -1.06382, -2.52995, -3.83095, -4.84728, -5.48249, -5.67212, -5.38976,
 -4.64977, -3.50645, -2.04964, -0.397182, 1.31508, 2.945, 4.3564, 5.43082, 6.07785, 6.25}
```

In[79]:=
```
y9 = Transpose[data91][[2]]
```

Out[79]=
```
{0., 0.499167, 0.993347, 1.4776, 1.94709, 2.39713, 2.82321, 3.22109, 3.58678, 3.91663,
 4.20735, 4.45604, 4.6602, 4.81779, 4.92725, 4.98747, 4.99787, 4.95832, 4.86924, 4.7315,
 4.54649, 4.31605, 4.04248, 3.72853, 3.37732, 2.99236, 2.57751, 2.1369, 1.67494, 1.19625,
 0.7056, 0.207903, -0.291871, -0.788728, -1.27771, -1.75392, -2.2126, -2.64918, -3.05929,
 -3.43883, -3.78401, -4.09139, -4.35788, -4.58083, -4.75801, -4.88765, -4.96846,
 -4.99962, -4.98082, -4.91226, -4.79462, -4.62907, -4.41727, -4.16134, -3.86382,
 -3.5277, -3.15633, -2.75343, -2.32301, -1.86938, -1.39708, -0.910813, -0.415447, 0.}
```

In[80]:=
```
L = Length[x9]
```

Out[80]=
```
64
```

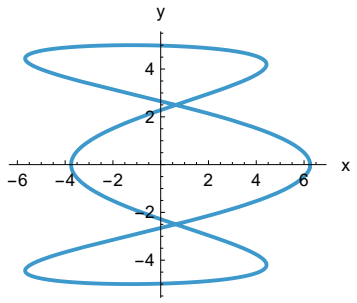
In[81]:=
```
{fx9 := Interpolation[x9, Method → "Spline"], fy9 := Interpolation[y9, Method → "Spline"]}
```

Out[81]=
```
{Null, Null}
```

In[82]:=
```
A9[t_] := {fx9[t], fy9[t]}
```

In[83]:=
```
tra9 = ParametricPlot[A9[t], {t, 1, L}]
```

Out[83]=



In[84]:=
```
nreg
```

Out[84]=

3

In[85]:=
```
colorList = RandomColor[10]
```

Out[85]=

{■, ■, ■, ■, ■, ■, ■, ■, ■, ■}

In[86]:=
```
ObjectNew = {Red, RegularPolygon[{0, 0}, RR5, nreg]}
```

Out[86]=

{■, RegularPolygon[{0, 0}, 5, 3]}

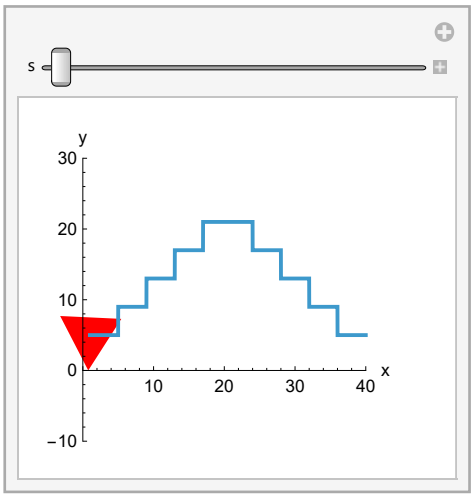
In[87]:=
```
ObjectRot[a_, s_] := GeometricTransformation[ObjectNew,
  Composition[TranslationTransform[fL5[s]], RotationTransform[a]]]
```

In[88]:=
```
GPnew[a_, s_] := Graphics[ObjectRot[a, s], PlotRange → {{0, 40}, {-10, 30}}, Axes → True]
```

In[89]:=
```
GGPnew[a_, s_] := Show[GPnew[a, s], Tra15]
```

In[90]:=
```
Manipulate[GGPnew[s, s], {s, 1, tmax}]
```

Out[90]=



In[91]:=
```
Clear[ObjectNew9]
```

In[92]:=
```
ObjectNew9[c_] := {c, RegularPolygon[{0, 0}, 2, nreg]}
```

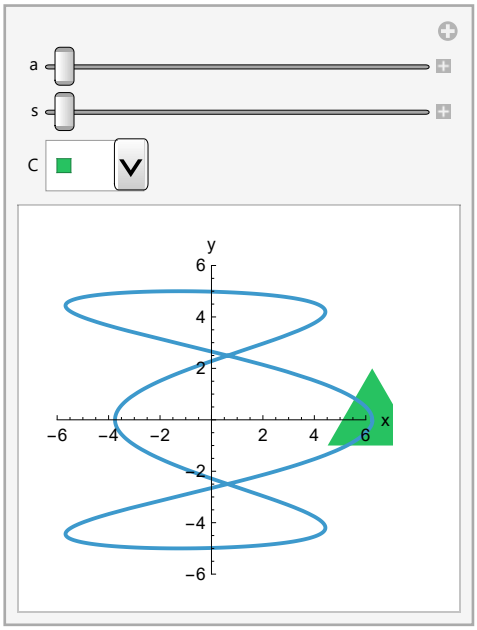
In[93]:=
```
ObjectRot9[a_, s_, c_] := GeometricTransformation[ObjectNew9[c],
  Composition[TranslationTransform[A9[s]], RotationTransform[a]]]
```

In[94]:=
```
GPnew9[a_, s_, c_] := Graphics[ObjectRot9[a, s, c], PlotRange → 6, Axes → True]
```

In[95]:=
```
ShowAll9[a_, s_, nreg_, C_] := Show[{GPnew9[a, s, C], tra9}]
```

In[96]:=
```
Manipulate[ShowAll9[a, s, nreg, C], {a, 0, 2 * Pi}, {s, 1, L}, {C, colorList}]
```

Out[96]=



In[97]:=
```
(* Problem 10 *)
```

In[98]:=
```
Clear[S1, etraj1]
```

In[99]:=
```
S1[u_, v_] := {u, v, u^2 + v^2}
```

In[100]:=
```
R1 = 2;
```

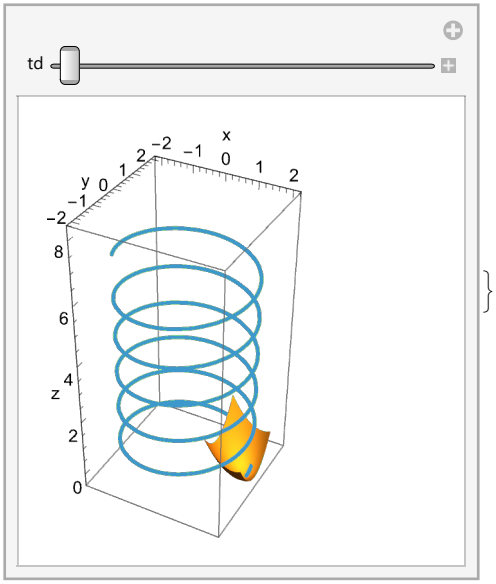
In[101]:=
```
etraj1[t_] := {R1 * Cos[5 * t], R1 * Sin[5 * t], t}
```

In[102]:=
```
MoveF0[F_, traj_, L0_] := Module[{t1, plottraj, SurfacePosF, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 0, L0}];
  SurfacePosF[u_, v_, t1_] := F[u, v] + traj[t1];
  plotF[t1_] :=
   ParametricPlot3D[SurfacePosF[u, v, t1], {u, -1, 1}, {v, -1, 1}, Mesh → None];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```

In[103]:=
```
L0 = 7;
```

In[104]:=
```
{M0 = MoveF0[S1, etraj1, L0]; Manipulate[M0[td], {td, 0, L0}]}
```

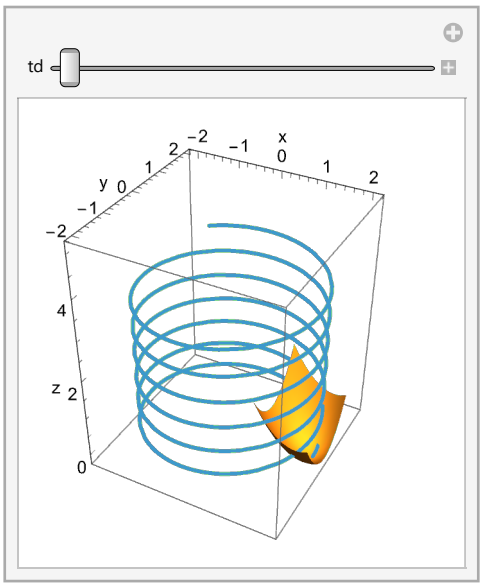
Out[104]=



In[105]:=

```
Manipulate[MoveF0[S1, etraj1, L0][td], {td, 0, 7}]
```

Out[105]=



In[106]:=

```
data10 = {{2.3511410091698925`, -3.23606797749979`, 3},
    {3.804226065180614`, 1.2360679774997898`, 3}, {0.`, 4.`, 3},
    {-3.804226065180614`, 1.2360679774997898`, 3}, {-2.3511410091698925`,
     -3.23606797749979`, 3}, {-2.3511410091698925`, -3.23606797749979`, 3}};
```

In[107]:=

```
L10 = Length[data10]
```

Out[107]=

6

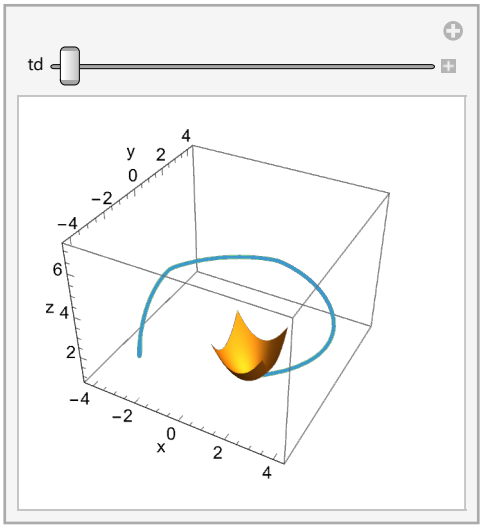
In[108]:=

```
Interp3D[data_] := Module[{x, y, z, fx, fy, fz},
  x = Transpose[data][[1]];
  y = Transpose[data][[2]];
  z = Transpose[data][[3]];
  fx = Interpolation[x];
  fy = Interpolation[y];
  fz = Interpolation[z];
  traj = {fx[#1], fy[#1], fz[#1]} &;
  traj]
```

In[109]:=

```
MoveF0New[F_, traj_, L0_] := Module[{t1, plottraj, SurfacePosF, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 1, L0}];
  SurfacePosF[u_, v_, t1_] := F[u, v] + traj[t1];
  plotF[t1_] :=
   ParametricPlot3D[SurfacePosF[u, v, t1], {u, -1, 1}, {v, -1, 1}, Mesh → None];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-4, 4}, {-4, 4}, {1, L0 + 1}}];
  plotF1]
```

In[110]:=
```
traj10 = Interp3D[data10]
```
Out[110]=
```
{fx$14150[⌗1], fy$14150[⌗1], fz$14150[⌗1]} &
```
In[111]:=
```
M10 = MoveF0New[S1, traj10, L10]
```
Out[111]=
```
plotF1$14151
```
In[112]:=
```
Manipulate[M10[td], {td, 1, L10}]
```
Out[112]=



In[113]:=
```
(* Songkran 1 *)
```
In[114]:=
```
Clear[S1]
```
In[115]:=
```
S1[u_, v_] := {u, v, u^2 + v^2}
```
In[116]:=
```
R1 = 2;
```
In[117]:=
```
etraj1[t_] := {R1 * Cos[10 * t], R1 * Sin[10 * t], t}
```
In[118]:=
```
Moveff0[F_, traj_, L0_] := Module[{t1, plottraj, SurfacePosF, plotpoints, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 1, L0}];
  SurfacePosF[u_, v_, t1_] := F[u, v] + traj[t1];
  plotF[t1_] :=
   ParametricPlot3D[SurfacePosF[u, v, t1], {u, -1, 1}, {v, -1, 1}, Mesh → None];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```
In[119]:=
```
(* Problem 1 *)
```

In[120]:=

```
L0 = 7
```

Out[120]=

```
7
```

In[121]:=

```
MoveF1[F_, traj_, L0_, ColorS_, ColorT_] :=
 Module[{t1, plottraj, SurfacePosF, plotpoints, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 1, L0}, PlotStyle → ColorT];
  SurfacePosF[u_, v_, t1_] := F[u, v] + traj[t1];
  plotF[t1_] := ParametricPlot3D[SurfacePosF[u, v, t1],
    {u, -1, 1}, {v, -1, 1}, Mesh → None, PlotStyle → ColorS];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```

In[122]:=

```
M1 = MoveF1[S1, etraj1, L0, Red, Purple]
```

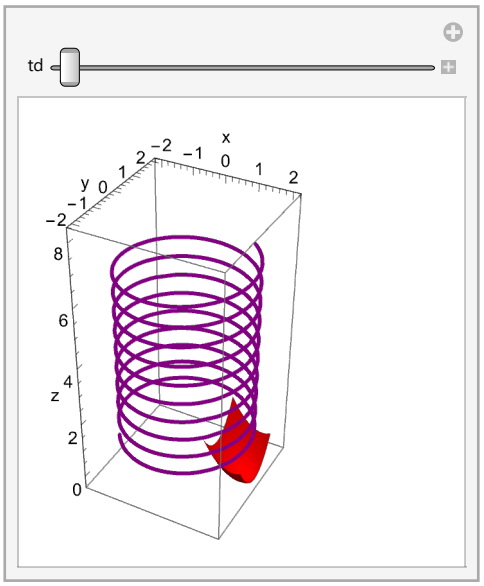
Out[122]=

```
plotF1$14643
```

In[123]:=

```
Manipulate[M1[td], {td, 0, L0, 0.001}]
```

Out[123]=



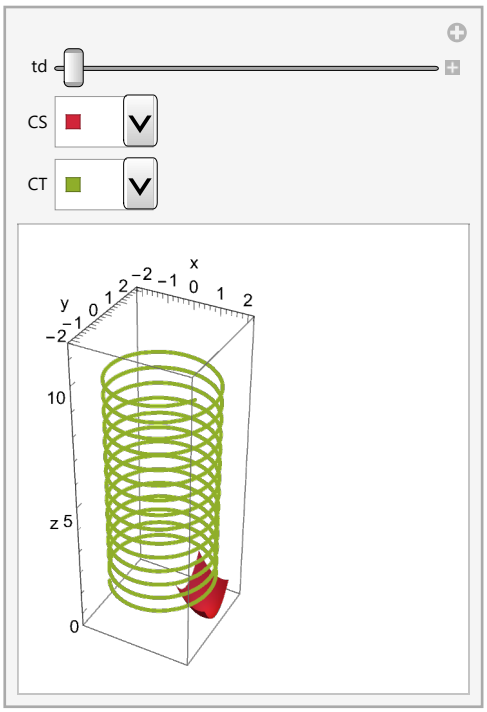
In[124]:=

```
L0 = 4; ListT = RandomColor[8]; ListS = RandomColor[8];
```

In[125]:=

```
Manipulate[MoveF1[S1, etraj1, 10, CS, CT][td], {td, 0, L0}, {CS, ListS}, {CT, ListT}]
```

Out[125]=



In[126]:=

```
(* Problem 2 *)
```

In[127]:=

```
L0 = 4;
```

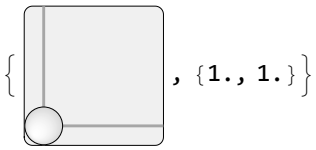
In[128]:=

```
MoveF2[F_, traj_, L0_, ColorS_, ColorT_, X_, Y_] :=
 Module[{t1, plottraj, SurfacePosF, plotpoints, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 1, L0}, PlotStyle → ColorT];
  SurfacePosF[u_, v_, t1_] := F[u, v] + traj[t1];
  plotF[t1_] := ParametricPlot3D[SurfacePosF[u, v, t1],
    {u, -1, 1}, {v, -1, 1}, Mesh → None, PlotStyle → ColorS];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-X, X}, {-Y, Y}, {0, L0 + 1}}];
  plotF1]
```

In[129]:=

```
SL1 = {Slider2D[Dynamic[xy], {{1, 1}, {10, 10}}], Dynamic[xy]}
```

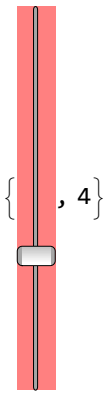
Out[129]=



In[130]:=

```
SL2 = {VerticalSlider[Dynamic[L0], {2, 8}, Background → Pink], Dynamic[L0]}
```

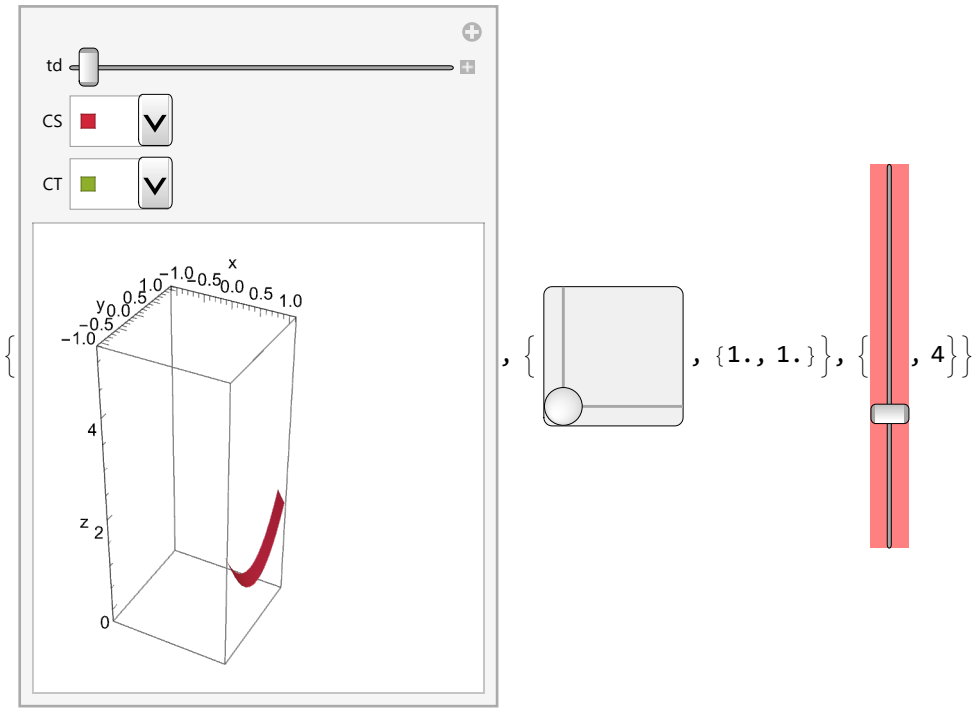
Out[130]=



In[131]:=

```
{Manipulate[MoveF2[S1, etraj1, L0, CS, CT, xy[[1]], xy[[2]]][td],
  {td, 0, L0}, {CS, ListS}, {CT, ListT}], SL1, SL2}
```

Out[131]=



In[132]:=

```
(* Problem 3 *)
```

In[133]:=

```
R1 = 2;
```

In[134]:=

```
etraj1[t_] := {R1 * Cos[10 * t], R1 * Sin[10 * t], t}
```

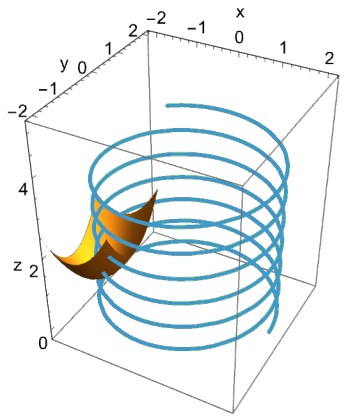
In[135]:=

```
Movef0[F_, traj_, L0_] := Module[{t1, plottraj, SurfacePosF, plotpoints, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 1, L0}];
  SurfacePosF[u_, v_, t1_] := F[u, v] + traj[t1];
  plotF[t1_] :=
   ParametricPlot3D[SurfacePosF[u, v, t1], {u, -1, 1}, {v, -1, 1}, Mesh → None];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```

In[136]:=

```
MoveF0[S1, etraj1, L0][1]
```

Out[136]=



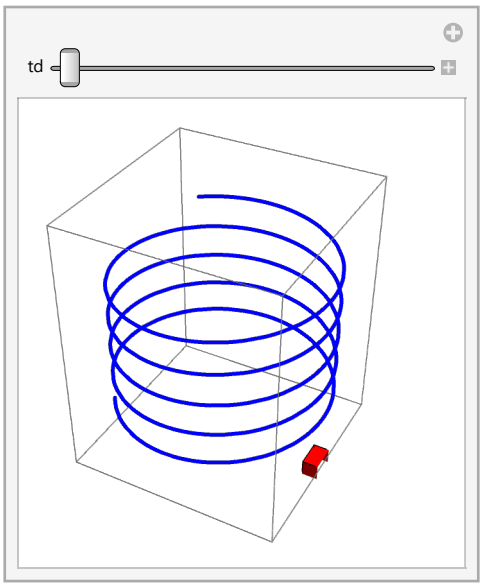
In[137]:=

```
MoveF3[F_, traj_, L0_, ColorS_, ColorT_, rr_] :=
 Module[{t1, plottraj, plotpoints, plotF1, plotF},
  plottraj = ParametricPlot3D[traj[t1], {t1, 1, L0}, PlotStyle → ColorT];
  plotF[t1_] := Graphics3D[{ColorS, F[traj[t1], rr]}];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```

In[138]:=

```
Manipulate[MoveF3[Cube, etraj1, L0, Red, Blue, 0.5][td], {td, 0, L0}]
```

Out[138]=



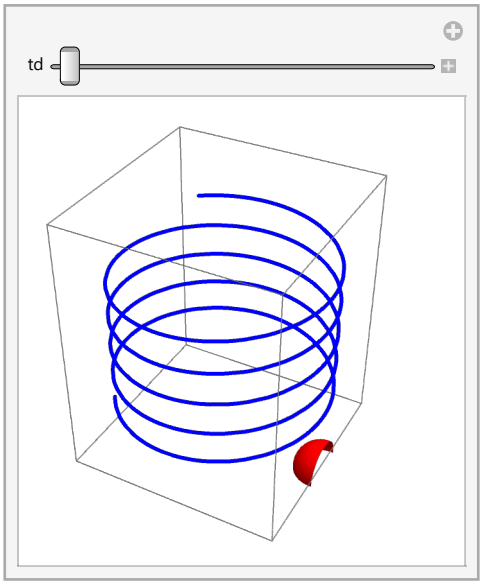
In[139]:=

```
(* Problem 4 *)
```

In[140]:=

```
Manipulate[MoveF3[Sphere, etraj1, L0, Red, Blue, 0.5][td], {td, 0, L0}]
```

Out[140]=



In[141]:=

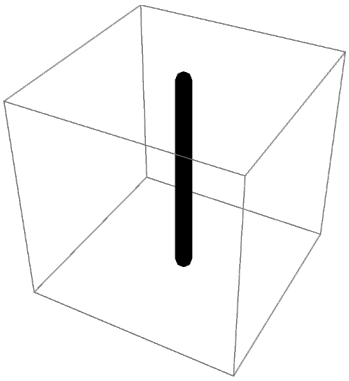
In[142]:=

```
LineR[P_, R_] := {Thickness[0.05], Line[{P - {0, 0, R}, P + {0, 0, R}}]}
```

In[143]:=

```
Graphics3D[LineR[{0, 0, 0}, 1]]
```

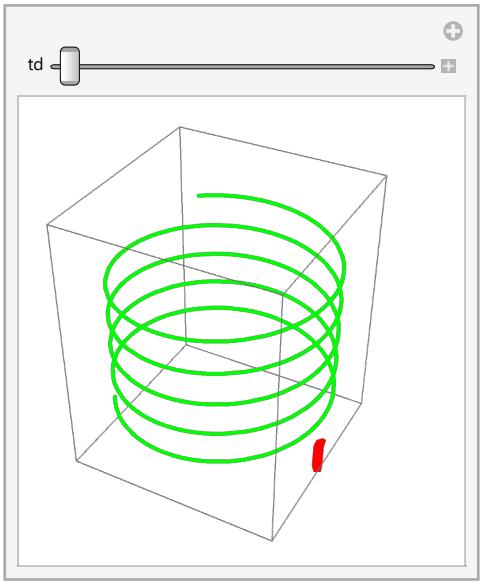
Out[143]=



In[144]:=

```
Manipulate[MoveF3[LineR, etraj1, L0, Red, Green, 0.5][td], {td, 0, L0}]
```

Out[144]=



In[145]:=

```
(* Problem 5.1 *)
```

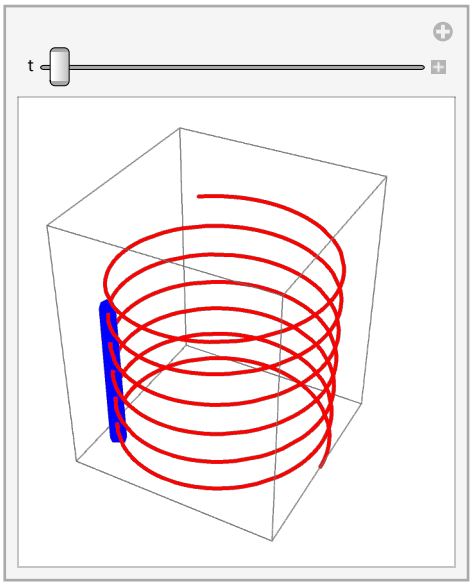
In[146]:=

```
MoveF5[F_, traj_, L0_, ColorS_, ColorT_, R1_] :=
 Module[{t1, plottraj, plotpoints, plotF1, plotF, Object, RandomColor},
  If[ColorS == ColorT, Return["Change the input colors"]];
  plottraj = ParametricPlot3D[traj[t1], {t1, 0, L0}, PlotStyle → ColorT];
  Object[t1_] := {ColorS, F[traj[t1], R1]};
  plotF[t1_] := Graphics3D[Object[t1], R1];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```

In[147]:=
```
MoveF5[Cube, etraj1, L0, Red, Red, 1.5]
```
Out[147]=
```
Change the input colors
```
In[148]:=
```
RandomColor[1]
```
Out[148]=
```
{■}
```
In[149]:=
```
{If[Red == Red, Blue, Red], F[traj[1], R1]}
```
Out[149]=
```
{■, F[{2.35114, -3.23607, 3}, 2]}
```
In[150]:=

In[151]:=
```
MoveF51[F_, traj_, L0_, ColorS_, ColorT_, R1_] :=
 Module[{plottraj, plotF1, plotF, Objectt, rc, rc1},
  rc = Red;
  rc1 = Blue;
  Objectt[t1_] := {If[ColorS == ColorT, rc = rc1, ColorS], F[traj[t1], R1]};
  plottraj = ParametricPlot3D[traj[t1], {t1, 0, L0}, PlotStyle → rc];


  plotF[t1_] := Graphics3D[Objectt[t1]];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```
In[152]:=
```
Manipulate[MoveF51[LineR, etraj1, L0, Blue, Blue, R1][t], {t, 1, L0}]
```
Out[152]=



In[153]:=
```
(* Problem 5.2 *)
```

In[154]:=
```
R2 = 0.8
```

Out[154]=
```
0.8
```

In[155]:=
```
MoveF52[F_, traj_, L0_, ColorS_, ColorT_, R1_] :=
 Module[{t1, SurfacePosF, plottraj, plotpoints, plotF1, plotF, Object},
  If[(! NumericQ[R1] || ! NumericQ[L0]), Return["R1 or L0 are not connected"]];
  plottraj = ParametricPlot3D[traj[t1], {t1, 0, L0}, PlotStyle → ColorT];
  Object[t1_] := {rc, F[traj[t1], R1]};
  plotF[t1_] := Graphics3D[Object[t1]];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → {{-2, 2}, {-2, 2}, {0, L0 + 1}}];
  plotF1]
```

In[156]:=
```
MoveF52[Cube, etraj1, L0, Red, Blue, S]
```

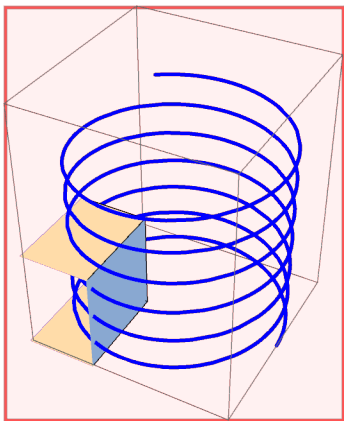
Out[156]=
```
R1 or L0 are not connected
```

In[157]:=
```
MoveF52[Cube, etraj1, S, Red, Blue, S]
```

Out[157]=
```
R1 or L0 are not connected
```

In[158]:=
```
MoveF52[Cube, etraj1, L0, Red, Blue, R1][1]
```

Out[158]=



In[159]:=
```
(* Problem 6 *)
```

In[160]:=

```
Interp3D[data_] := Module[{x, y, z, fx, fy, fz},
  x = Transpose[data][[1]];
  y = Transpose[data][[2]];
  z = Transpose[data][[3]];
  fx = Interpolation[x];
  fy = Interpolation[y];
  fz = Interpolation[z];
  traj = {fx[#1], fy[#1], fz[#1]} &;
  traj]
```

In[161]:=

```
data6 = {{-4, 2, 4}, {4, 5, -1}, {2, 5, 0}, {1, 1, 3}, {2, -6, -6},
    {-4, -4, -4}, {0, 0, 0}, {0, 0, 0}, {1, 5, 0}, {1, 1, 3}, {1, -1, -1}, {2, 2, 2}};
```

In[162]:=

```
L6 = Dimensions[data6][[1]]
```

Out[162]=

```
12
```

In[163]:=

```
Interp3D[data6][1]
```

Out[163]=

```
{-4, 2, 4}
```

In[164]:=

```
MoveF6[F_, data_, ColorS_, ColorT_, R1_] :=
 Module[{t1, SurfacePosF, plottraj, plotpoints, plotF1, plotF, Object},
  plottraj = ParametricPlot3D[Interp3D[data6][t1], {t1, 1, L6}, PlotStyle → ColorT];
  Object[t1_] := {Red, F[traj[t1], R1]};
  plotF[t1_] := Graphics3D[Object[t1]];
  plotF1[t1_] := Show[{plotF[t1], plottraj}, PlotRange → 7];
  plotF1]
```

In[165]:=

```
Manipulate[MoveF6[Sphere, data6, Red, Blue, 1.5][td], {td, 1, L6}]
```

Out[165]=

